

Qu Question 1:

$N = P = 0.5$  (success)

K:  $X =$  Number of companies giving 4 weeks of vacation after 15 years

$X$  Number of trial =  $n$

a  $n = 6$

$$\begin{aligned} \text{a) } P(2 \leq X \leq 5) &= \sum_{x=2}^5 \binom{6}{x} (0.5)^x (0.5)^{6-x} \\ &= P(2) + P(3) + P(4) + P(5) \\ &= \binom{6}{2} (0.5)^2 + \binom{6}{3} (0.5)^3 + \binom{6}{4} (0.5)^4 \\ &\quad + \binom{6}{5} (0.5)^5 \\ &= \frac{1}{64} (15 + 20 + 15 + 6) = \frac{56}{64} \end{aligned}$$

$$\text{b) } P(X < 3) = 0.875.$$

$$\begin{aligned} \text{a) } &= P(0) + P(1) + P(2) = \binom{6}{0} (0.5)^6 + \binom{6}{1} (0.5)^5 \\ &\quad + \binom{6}{2} (0.5)^4 \\ &= \frac{1 + 6 + 15}{64} = \frac{22}{64} = 0.34375. \end{aligned}$$

Question 2:  $p = 0.3$   $n = 10$

$$\begin{aligned} \text{a) } P(X=3) &= \binom{10}{3} (0.3)^3 (1-0.3)^7 \\ &= 0.2668 \end{aligned}$$

$$\begin{aligned} \text{b) } P(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - \sum_{x=0}^3 \binom{10}{x} (0.3)^x (0.7)^{10-x} \\ &= 1 - (P(0) + P(1) + P(2) + P(3)) \\ &= 1 - (0.2668 + \binom{10}{0} (0.3)^0 (0.7)^{10} + \binom{10}{1} (0.3)^1 (0.7)^9 \\ &\quad + \binom{10}{2} (0.3)^2 (0.7)^8) \\ &= 0.3504. \end{aligned}$$

nal Distribution

Question 3:

$$p=0.25 \quad n=15$$

$$(a) P(3 \leq x \leq 6)$$

$$\begin{aligned} P(3 \leq x \leq 6) &= \sum_{x=3}^6 \binom{15}{x} (0.25)^x (1-0.25)^{15-x} \\ &= P(3) + P(4) + P(5) + P(6) \\ &= {}^{15}C_3 (0.25)^3 (0.75)^{12} + {}^{15}C_4 (0.25)^4 (0.75)^{11} + {}^{15}C_5 (0.25)^5 (0.75)^{10} \\ &\quad + {}^{15}C_6 (0.25)^6 (0.75)^9 \\ &= 0.2252 + 0.2252 + 0.1651 + 0.0917 \end{aligned}$$

$$P = 0.7073$$

$$\begin{aligned} b) P(x < 4) &= P(0) + P(1) + P(2) + P(3) \\ &= 0.0134 + 0.0668 + 0.1539 + 0.2552 \end{aligned}$$

$$P(x < 4) = 0.4613$$

$$c) P(x \geq 5) = \sum_{x=5}^{15} = 0.566$$

Question 4:  $n=8$   $p=0.6$

$$P(x=6) = {}^8C_6 (0.6)^6 (0.4)^2 = 0.2090$$

Question 5:

Total missiles = 10

defective = 3

selected = 4

$N=10$

$K=$

a) All 4 fire

$$P = \frac{{}^7C_4 \times {}^3C_0}{{}^{10}C_4} = 0.1667$$

$$b) P(3) = \frac{{}^7C_3 {}^4C_1}{{}^{10}C_4} = 0.5$$

$$P(2) = \frac{{}^7C_2 {}^3C_2}{{}^{10}C_4} = 0.3$$

$$P(\text{at most 2 do not fire}) = 0.1667 + 0.5 + 0.3 = 0.9667$$

# Question 6 (Hypergeometric Distribution)

$$N=20 \quad \text{selected} = 5$$

$$K=3 \quad n=5$$

$X$  = number of voting firms.

$$a) P(X=0) = \frac{{}^3C_0 {}^{17}C_5}{{}^{20}C_5} = 0.3991$$

$$b) P(X=2) = \frac{{}^3C_2 {}^{17}C_3}{{}^{20}C_5} = 0.1316$$

# Question 7: (Poisson Distribution)

1. fixed Interval
2. Randomly and Independently
3. mean / average is given

$$P(X=K) = \frac{e^{-\lambda} \lambda^K}{K!}$$

$$\lambda = 2$$

$$a) P(X \geq 4) = 1 - P(X < 4)$$

$$= 1 - (P(0) + P(1) + P(2) + P(3))$$

$$= 1 - \left[ \left( \frac{e^{-2} 2^0}{0!} \right) + \left( \frac{e^{-2} 2^1}{1!} \right) + \left( \frac{e^{-2} 2^2}{2!} \right) + \left( \frac{e^{-2} 2^3}{3!} \right) \right]$$

$$= 1 - [0.1353 + 0.2707 + 0.2707 + 0.1804]$$

$$= 1 - 0.8571 = 0.1429.$$

$$b) P(X=0) = \frac{e^{-2} 2^0}{0!} = e^{-2} = 0.1353.$$



Question 8: Continuous Uniform distribution  
 $a = 7$   $b = 10$

PDF  $f(x) = \frac{1}{b-a} = \frac{1}{3}$

for  $7 \leq x \leq 10$

CDF:

$$P(x_1 < X < x_2) = \frac{x_2 - x_1}{b - a}$$

$$a) P(X \leq 8.8) = \frac{8.8 - 7}{10 - 7} = \frac{1.8}{3} = 0.6$$

$$b) P(7.4 < X < 9.5) = \frac{9.5 - 7.4}{3} = 0.70$$

$$c) P(X \geq 8.5) = \frac{10 - 8.5}{3} = 0.5$$

Question 9: Normal Distribution

$$\mu = 40 \text{ month}$$

$$\sigma = 6.3 \text{ months}$$

$$a) Z = \frac{X - \mu}{\sigma}$$

$$P(X > 32) = \frac{32 - 40}{6.3} = -1.27$$

$$P(X > 32) = P(Z > -1.27)$$
$$= 1 - P(Z < -1.27)$$
$$= 1 - 0.1021$$

$$= 0.8979$$

$$b) P(X < 28) = P(Z < \frac{28 - 40}{6.3}) = P(Z < -1.90)$$

$$= 0.0284$$

$$P(Z > -0.48) = 0.315$$

$$c) P(37 < X < 49)$$

$$Z = \frac{37 - 40}{6.3} = -0.48$$

$$Z = \frac{49 - 40}{6.3} = 0.9215$$

$$P(37 < X < 49) = 0.9215 - 0.3150 = 0.6065$$

10. Normal Distribution

$$\mu = 8 \text{ kg}, \quad \sigma = 0.9 \text{ kg}$$

a) Weight over 9.5 kg  $z = \frac{9.5 - 8}{0.9} = 1.67$

$$\begin{aligned} P(X > 9.5) &= 1 - P(Z > 1.67) \\ &= 1 - P(Z > 1.67) \\ &= 1 - 0.9552 \\ &= 0.0448. \end{aligned}$$

b)  $P(X \leq 8.6) = P(Z \leq 0.67)$

$$z = \frac{8.6 - 8}{0.9} = 0.67$$

$$= 0.7475$$

c)  $P(7.3 \leq X \leq 9.1)$

$$z = \frac{7.3 - 8}{0.9} = -0.78$$

$$z = \frac{9.1 - 8}{0.9} = 1.22$$

$$P(Z < 1.22) = 0.8880$$

$$P(Z < -0.78) = 0.2172$$

$$\begin{aligned} P(7.3 \leq X \leq 9.1) &= 0.888 - 0.2172 \\ &= 0.6708. \end{aligned}$$

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